KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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**Course Code : EE 4121**

# VLSI

## 2.7 NMOS Inverter with a Resistor Load

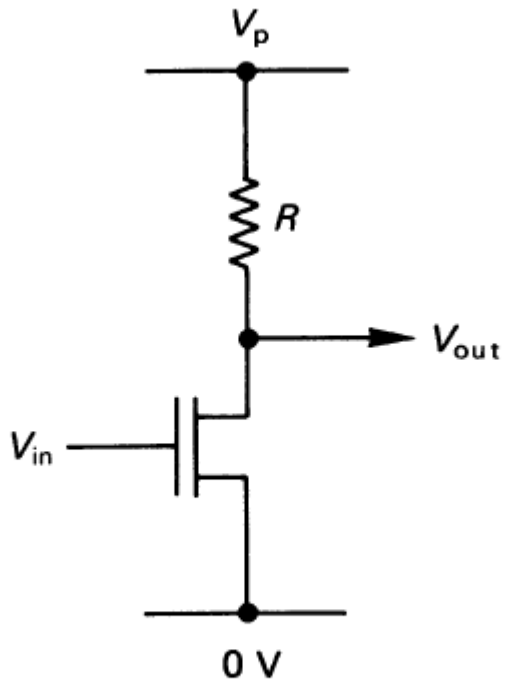



*Figure 2.11* NMOS inverter with a resistor load

In order to see what value of $R$ is required, it is necessary to take typical values for the parameters. $V_p$ will be taken as 5 V since most MOS and bipolar logic families operate at this voltage. $\epsilon\mu_n/D$ will be taken as 30 $\mu$A/V$^2$ and $V_{te}$ as 1 V. An equal gate width and length are assumed, so the width-to-length ratio of the transistor, known as the 'aspect ratio', is 1/1.

$R$ is calculated to give a low $V_{out}$ of 0.3 $V_{te}$, since this gives a reasonable noise margin between $V_{out}$ and the gate threshold. Thus $V_{ds}$ of the transistor is 0.3 V and, since $V_{gs}$ is 5 V, $V_{ds} < V_{gs} - V_{te}$ and the device is operating in the resistive region. Equation (2.2) can now be used to find that

$$I_{ds} = 30\left(\frac{1}{1}\right)\left[(5-1)\,0.3 - \frac{0.3^2}{2}\right] = 34.7\ \mu\text{A}$$

This current also flows through the resistor. Hence

$$R = \frac{V_p - V_{out}}{I_{ds}} = \frac{5 - 0.3\ \text{V}}{34.7\ \mu\text{A}} = 135.4\ \text{k}\Omega$$

The silicon area required to implement this resistor is far larger than that for the transistor (typically by a factor of 300). Thus it is not practical to use a resistor as a load and so an MOS device is used instead.

## 2.8 NMOS Inverter with an NMOS Enhancement Transistor Load

This section considers the use of the same transistor type for both the load and the digital switch, see figure 2.12. T1 acts as the switch and is also referred to as the 'driver'. T2 is the load and its gate is connected to $V_p$ in order to maximise its $V_{gs}$. Both transistors have the same threshold $V_{te}$.

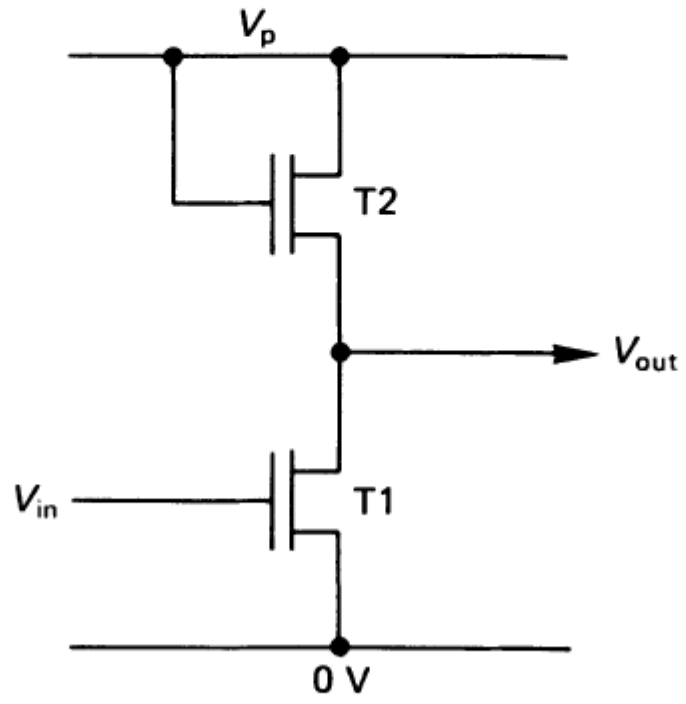



*Figure 2.12* NMOS inverter with NMOS enhancement load

When $V_{in}$ is low, it is less than $V_{te}$ so transistor T1 is off. However, some very small leakage current flows through T1 and this is supplied by T2. Thus T2 is just brought to the edge of conduction and $V_{gs} - V_{te}$ for T2 is approximately 0 V, so

$$V_{out} = V_p - V_{te}$$

Again, taking $V_p$ as 5 V and $V_{te}$ as 1 V, the high level output voltage is 4 V if the body effect of T2 is neglected. However, T2's source-substrate voltage is significant at this magnitude of output voltage and the effect on the threshold voltage cannot be ignored. Using the expression for modifying the threshold voltage (given in section 2.3) with a substrate voltage of 0 V

$$V_{out} = V_p - V_{te} = V_p - [V_{te0} + \gamma(V_{sb})^{1/2}] = 5 - [1 + 0.5(V_{out})^{1/2}]$$

where the body effect constant $\gamma$ has been taken as 0.5. Rearranging yields

$$(V_{out})^{1/2} = 8 - 2V_{out}$$

By squaring each side and solving the resulting quadratic equation, the high level $V_{out}$ is found to be only 3.12 V.

The output from a gate is normally used to drive the input of other gates. Hence the high level input voltage for a gate is 3.12 V. When $V_{in}$ is high at this level, it exceeds T1's threshold so T1 is on and conducts current. This current is supplied by T2 which is also on. The circuit output is required to be low for a high input and this can be obtained by selecting suitable widths and lengths for the gates of the driver and the load transistors. Again, it is appropriate to choose the gate sizes so that the low level output is 0.3 $V_{te}$.

Consider T1 with a $V_{in}$ of 3.12 V and a $V_{out}$ of 0.3 V. Since T1's source is at 0 V, its $V_{gs}$ is 3.12 V and its $V_{ds}$ is 0.3 V. Thus $V_{ds} < V_{gs} - V_{te}$ and so T1 operates in the resistive region. Applying equation (2.2) with an $\epsilon\mu_n/D$ of 30 $\mu$A/V$^2$

$$I_{ds} \text{ of T1} = 30\left(\frac{W_1}{L_1}\right)\left[(3.12 - 1)0.3 - \frac{0.3^2}{2}\right] = 17.73\left(\frac{W_1}{L_1}\right)\ \mu\text{A}$$

where $(W_1/L_1)$ is the aspect ratio of transistor T1.

T2 is on and $V_{ds} = V_{gs}$. Hence the inequality $V_{ds} > V_{gs} - V_{te}$ is true and T2 is saturated. The gate-source voltage of T2 is $V_p - V_{out}$ and since $V_{out}$ is low, the body effect can be neglected. Applying equation (2.3) with a $V_{gs}$ of 4.7 V yields

$$I_{ds} \text{ of T2} = \frac{30}{2}\left(\frac{W_2}{L_2}\right)(4.7 - 1)^2 = 205.4\left(\frac{W_2}{L_2}\right)\mu\text{A}$$

where $(W_2/L_2)$ is the aspect ratio of T2. $I_{ds}$ of T1 and T2 are equated to obtain an inverter ratio $k$ for the circuit

$$k = \frac{(W_1/L_1)}{(W_2/L_2)} = 11.6$$

It is usual to split the inverter ratio obtained between transistors T1 and T2 in order to prevent T1 from occupying a much larger area than T2. Thus a ratio of 11.6 would be taken as 12 for convenience and could be split as $(W_1/L_1) = 3/1$ and $(W_2/L_2) = 1/4$.

The main drawback of an NMOS enhancement load, apart from the loss of voltage for high outputs, is the speed of the rising edge of the gate. This can be appreciated by considering that there is some capacitance $C_{out}$ on $V_{out}$, and $V_{out}$ is at 0.3 V corresponding to a $V_{in}$ of 3.12 V. If $V_{in}$ now switches to 0.3 V, T1 turns off and $V_{out}$ starts to rise. Initially, $V_{gs}$ of T2 is 4.7 V but as $V_{out}$ rises, $V_{gs}$ of T2 decreases and T2's threshold rises (because of the body effect) so that less current flows in T2. This progressively slows down the rate of charging up $C_{out}$, and as $V_{out}$ approaches 3.12 V, T2 tends to turn off and very little current flows to finish the charging of $C_{out}$. For this reason, n-channel depletion devices are preferred as a load.

## 2.9 NMOS Inverter with an NMOS Depletion Transistor Load

Referring to figure 2.13, the NMOS depletion transistor T2 forms the circuit load. The depletion threshold $V_{td}$ of T2 is negative and since the gate-source voltage of T2 is zero, T2 is always on.

When $V_{in}$ is low, it is less than the enhancement threshold $V_{te}$ of T1, and T1 is off. T2 is on and supplies the very small leakage current of T1. Since $I_{ds}$ of T2 is negligible, the drain-source voltage of T2 is very small, leading to a $V_{out}$ of $V_p$. Thus there is no high level output voltage loss as with the NMOS enhancement load.

The high level input voltage to a gate is therefore $V_p$. This exceeds T1's threshold voltage and hence T1 and T2 are both on. Again it is necessary to choose the gate dimensions of the transistors in order to obtain the desired low level $V_{out}$. Taking $V_{out}$ as $0.3V_{te}$ and $V_p$ as 5 V again, T1's $V_{gs}$ is 5 V and its $V_{ds}$ is 0.3 V. Thus $V_{ds} < V_{gs} - V_{te}$ and T1 is operating in the resistive region. Equation (2.2) is applied to express $I_{ds}$ of T1 in terms of its aspect ratio.

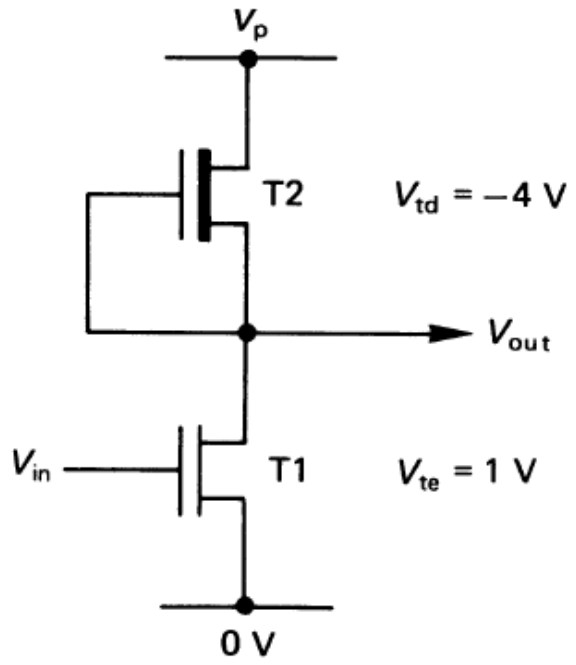



*Figure 2.13* NMOS inverter with NMOS depletion load

$$I_{ds} \text{ of T1} = 30\left(\frac{W_1}{L_1}\right)\left[(5-1)\,0.3 - \frac{0.3^2}{2}\right] = 34.65\left(\frac{W_1}{L_1}\right)\mu\text{A}$$

T2 is on with a $V_{gs}$ of 0 V and a $V_{ds}$ of $V_p - V_{out} = 4.7$ V. Thus $V_{ds} > V_{gs} - V_{td}$ and T2 is saturated. Equation (2.3) is applied to find T2's $I_{ds}$

$$I_{ds} \text{ of T2} = \frac{25}{2}\left(\frac{W_2}{L_2}\right)[0-(-4)]^2 = 200\left(\frac{W_2}{L_2}\right)\mu\text{A}$$

Note that $\epsilon\mu_n/D$ for a depletion transistor is typically 25 $\mu A/V^2$ compared with 30 $\mu A/V^2$ for an NMOS enhancement device (owing to the fact that a depletion transistor requires additional impurity doping to define a threshold of $-4$ V). Equating the current in T1 and T2 yields

$$k = \frac{(W_1/L_1)}{(W_2/L_2)} = 5.8$$

For convenience, this inverter ratio would be taken as 6 and split as $(W_1/L_1) = 3/1$ and $(W_2/L_2) = 1/2$.

It is possible to reduce the inverter ratio and thus the silicon area occupied by T1 and T2 at the expense of a reduced noise margin. The lower limit for $k$ is generally accepted to be 4 and this allows 0.5 V between the low level $V_{out}$ and the enhancement threshold. However, the author recommends that a value of $k = 6$ or greater be adopted for practical designs.

## 2.10 Edge Times for NMOS Inverter with a Depletion Load

The circuit of figure 2.13 represents a practical inverter that can be implemented in NMOS technology. It is thus worth examining the edge time response of the gate, as this will indicate the maximum speed of operation and the factors upon which the speed is dependent.

In the circuit of figure 2.13 the maximum current capability of T1 and T2 is different. This can be seen by considering the saturation currents of the devices, and it has already been shown that this current for T2 is $200(W_2/L_2)\,\mu$A. The maximum current through T1 occurs when the gate-source voltage is a maximum (5 V) and the device is saturated. Using equation (2.3)

$$\text{maximum } I_{ds} \text{ of T1} = \frac{30}{2}\left(\frac{W_1}{L_1}\right)(5-1)^2 = 240\left(\frac{W_1}{L_1}\right)\mu\text{A}$$

Thus

$$\frac{\text{maximum } I_{ds} \text{ of T1}}{\text{maximum } I_{ds} \text{ of T2}} = \frac{6(W_1/L_1)}{5(W_2/L_2)} = \frac{6k}{5}$$

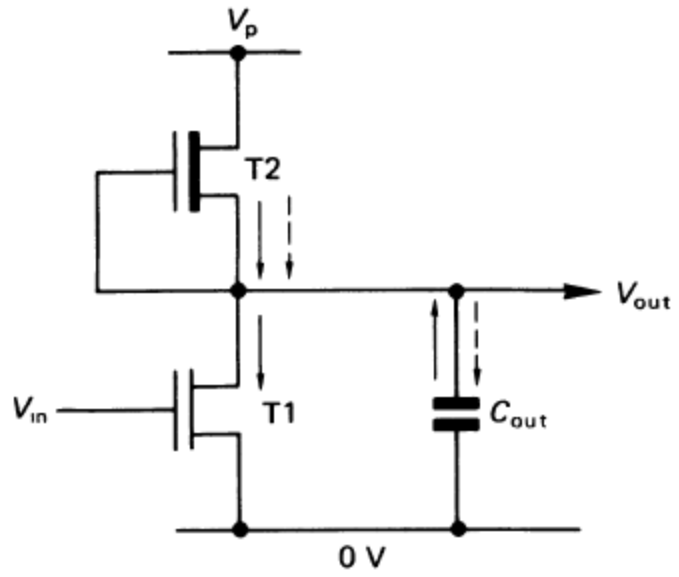



*Figure 2.14* Current flow during edge times: ↓ rising edge current flow, ↓ current flow during falling edge

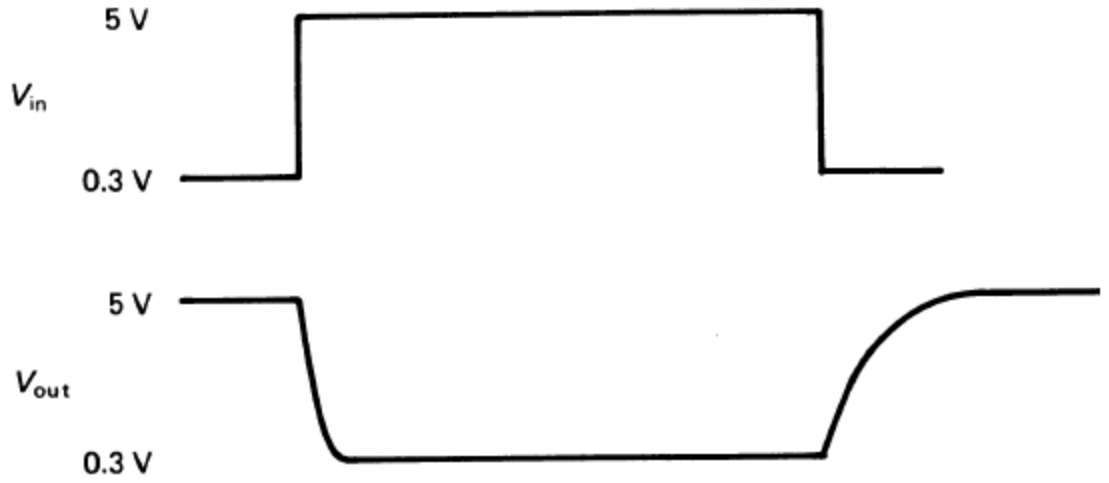



*Figure 2.15* Output response to an input step

Consider that the input is high and that the output has settled to its low level value of 0.3 V. If the input now changes instantaneously from high to low, then T1 turns off. T2 remains on and supplies current to charge the load capacitance (see figure 2.14). This causes the output to gradually rise to 5 V. The current $I_c$ in the capacitor is related to the change in voltage $dV_{out}$ across it with respect to time $dt$ by

$$I_c = C_{out} \frac{dV_{out}}{dt}$$

The rise time is dependent upon the rate at which the capacitance can be charged during this period. This in turn depends upon the current available from T2.

When the input changes instantaneously from low to high, T1 turns on. Both T1 and T2 are on and during the fall time, T1 accepts current from T2 and from $C_{out}$ as it discharges (see figure 2.14). Since the current in T2 is small compared with that in T1, the current T1 can accept determines the rate at which the capacitor discharges and hence the fall time.

It can therefore be expected that the falling and rising edge times will differ by a factor equal to the ratio of the current capability of T1 and T2. On this basis, the rising edge time will be $6k/5$ times that of the falling edge time. This is illustrated in figure 2.15.

Consider in detail the rising edge at $V_{out}$. The current supplied by T2 causes the output to rise from 0.3 V to 5 V, as shown in figure 2.16a. It is usual to calculate edge times from the 10 per cent to 90 per cent points on the output waveform, as many edges are exponential in shape and take a disproportionately long time to settle to their final value. Applying this to the depletion load inverter, the rise time is the time for the output to rise from 0.8 V to 4.5 V.

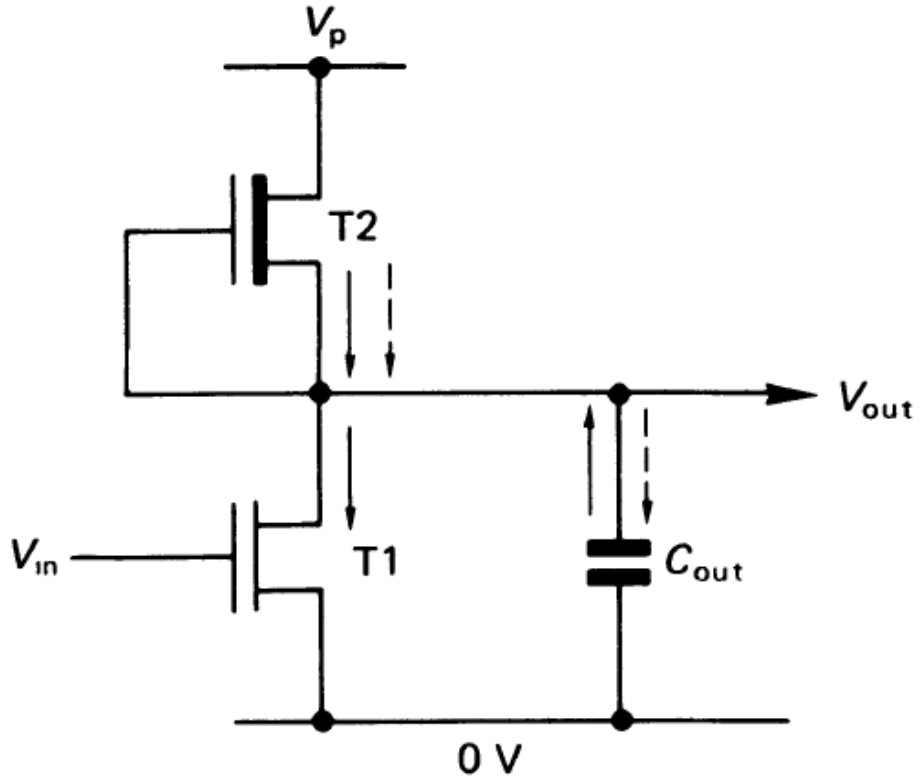
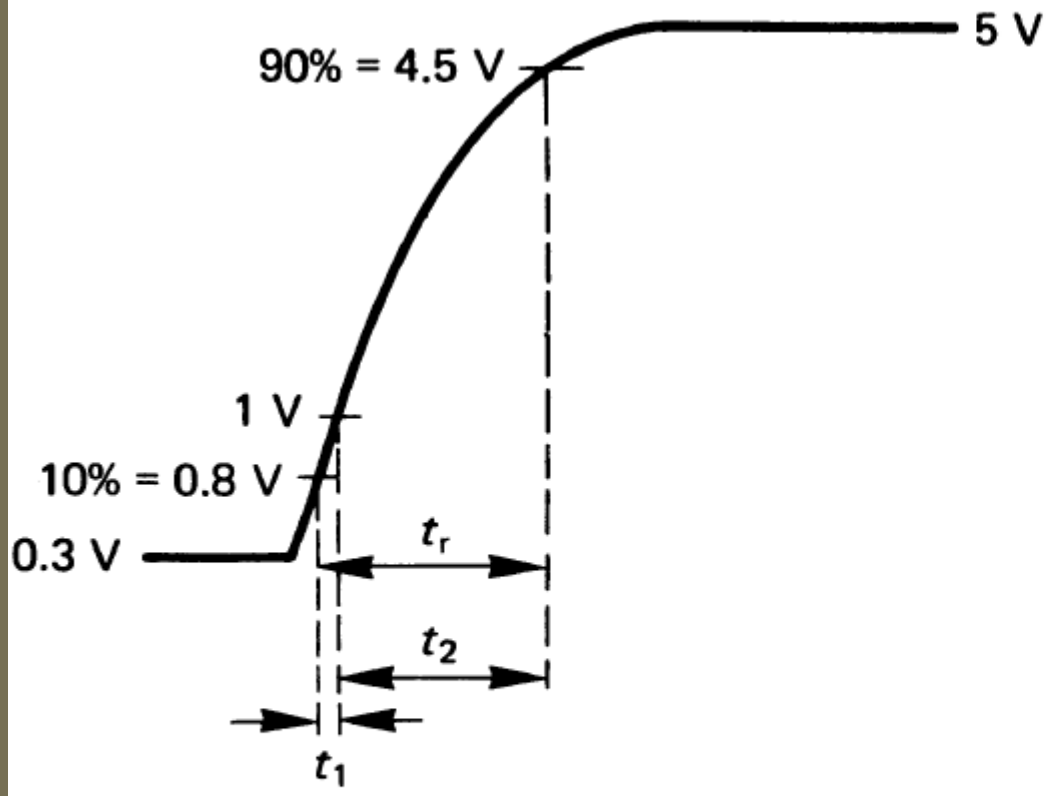
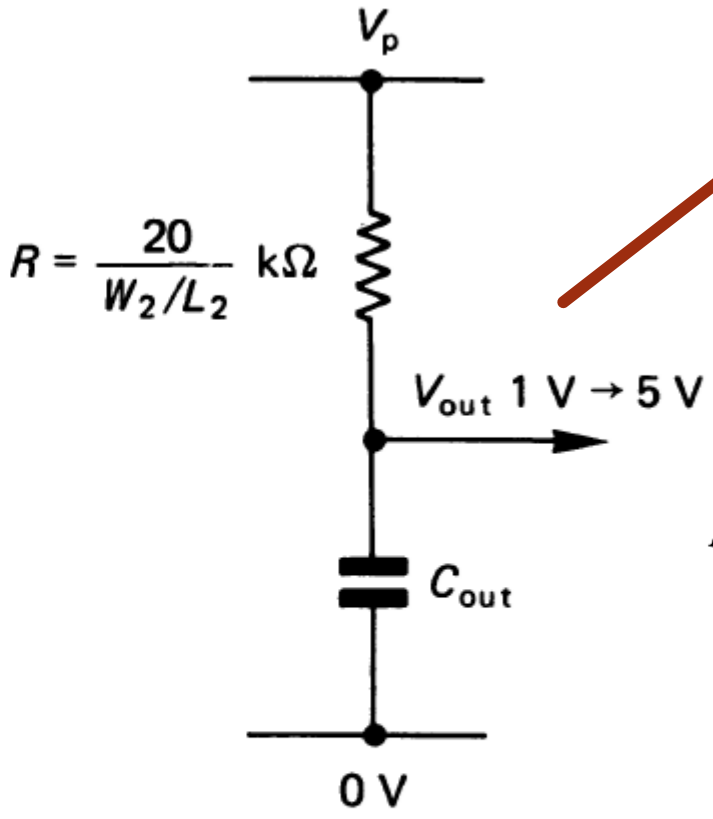



*Figure 2.14* Current flow during edge times: ↓ rising edge current flow, ↓ current flow during falling edge




(a)




(b)

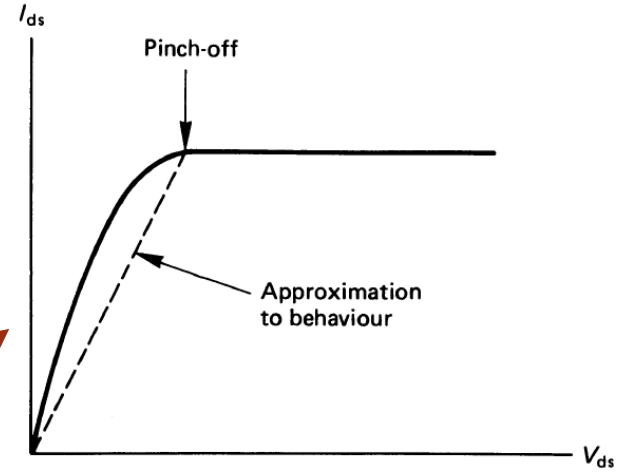



*Figure 2.17* Characteristic approximation

$$R = \frac{\text{pinch-off voltage}}{\text{pinch-off current}} = \frac{(V_{gs} - V_t)}{\frac{\epsilon\mu_n}{2D}\left(\frac{W}{L}\right)(V_{gs} - V_t)^2}$$

$$= \frac{1}{\frac{\epsilon\mu_n}{2D}\left(\frac{W}{L}\right)(V_{gs} - V_t)}$$

$$R = \frac{1}{\frac{25}{2}(W_2/L_2)[0 - (-4)]} \text{ M}\Omega = \frac{20}{(W_2/L_2)} \text{ k}\Omega$$

*Figure 2.16* Rise time behaviour: (a) rising edge, (b) circuit approximation

Thus the equivalent circuit during time $t_2$ simplifies to an $RC$ network, as shown in figure 2.16b. The output voltage $V_{out}$ across such a network, where the initial voltage across the capacitor is $V_i$ and the final voltage is $V_p$, is given by

$$V_{out} = V_p - (V_p - V_i) \exp[-t/(RC_{out})]$$

The time for the output to reach 4.5 V with an initial output of 1 V is

$$4.5 = 5 - (5 - 1)\exp[-t_2/(RC_{out})]$$

giving

$$t_2 = 2.08RC_{out}$$

If $C_{out}$ is in picofarads and $R$ in kilohms, then $t_2$ is given in nanoseconds. Substituting for T2's value of $R$, the rise time is

$$t_r = t_2 = \frac{42 C_{out}}{(W_2/L_2)} \text{ ns}$$

From previous considerations concerning the current drive capability of T1 relative to T2, the fall time $t_f$ is

$$t_f = \frac{5t_r}{6k} = \frac{5(W_2/L_2)42C_{out}}{6(W_1/L_1)(W_2/L_2)}$$

$$t_f = \frac{35 C_{out}}{(W_1/L_1)} \text{ ns}$$

The expression for the fall time agrees well with results obtained by simulating the inverter circuit. However, the expression for the rise time is not so accurate as the body effect is significant for T2. As the output rises, the threshold of T2 rises, reducing the current in T2. This increases the time necessary to charge the output capacitance. Simulation results for the inverter show that

$$t_r = \frac{60 C_{out}}{(W_2/L_2)} \text{ ns}$$

is more appropriate for the rise time.

In a fabrication process where the minimum allowable gate width or length is 6 $\mu$m, an inverter having T1 and T2 aspect ratios of 3/1 and 1/2 respectively will have a typical input capacitance of 0.05 pF and an output capacitance of 0.1 pF. Thus the circuit exhibits a rise time of 12 ns and a fall time of 1.2 ns.

The speed at which the circuit operates is dominated by the load capacitance. In practice, the output of a circuit drives other inputs and each input causes an increase in the load capacitance and hence the edge times. The absolute maximum frequency (in hertz) at which a circuit can operate is $1/(t_r + t_f)$ so, taking the capacitance figures of the last example, an inverter driving a similar inverter has a total load capacitance of 0.15 pF and an upper frequency limit of 50 MHz.